

Reproducing the Tables and Choosing the Integrator

Based on A. Lima-Silva and F. D. Freitas, "Exploring a Dynamic Homotopy Technique to Enhance the Convergence of Classical Power Flow Iterative Solvers in Ill-Conditioned Power System Models", *Energies* 17(18):4642, 2024

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CSE 402 Numerical Analysis Sessional

My part of the project

1

Reproduce Tables 2 to 5

FE, RK2 and BE along the path, our value next to the paper's

2

Figure 3 and Table 6

five-point paths and a fair timing benchmark

3

Other step rules

RK4 and BE-chord: does a better integrator help?

4

Tune δ

golden-section search for the best shift on every grid

Four ways to take a step along the path

$$\dot{x} = f(x, t) = -G_x(x, t)^{-1} G_t(x)$$

Scheme	Step	LUs	Type
Forward Euler	$x_{k+1} = x_k + \Delta t f(x_k, t_k)$	1	explicit
Backward Euler (paper)	$x_{k+1} = x_k + \Delta t f(x_k, t_{k+1})$	1	implicit
RK2	average of 2 slopes	2	explicit
RK4 (added)	weighted average of 4 slopes	4	explicit

Every slope costs one sparse LU, so cost = number of LUs.

Why backward Euler wins

FE overshoots

$$\Delta x_{FE} \approx \frac{t_{k+1}}{t_k} \Delta x_{BE}$$

- FE uses G_x at the old time, about t_k J: far too small
- **step 0.01 → 0.51: about 51× too long**

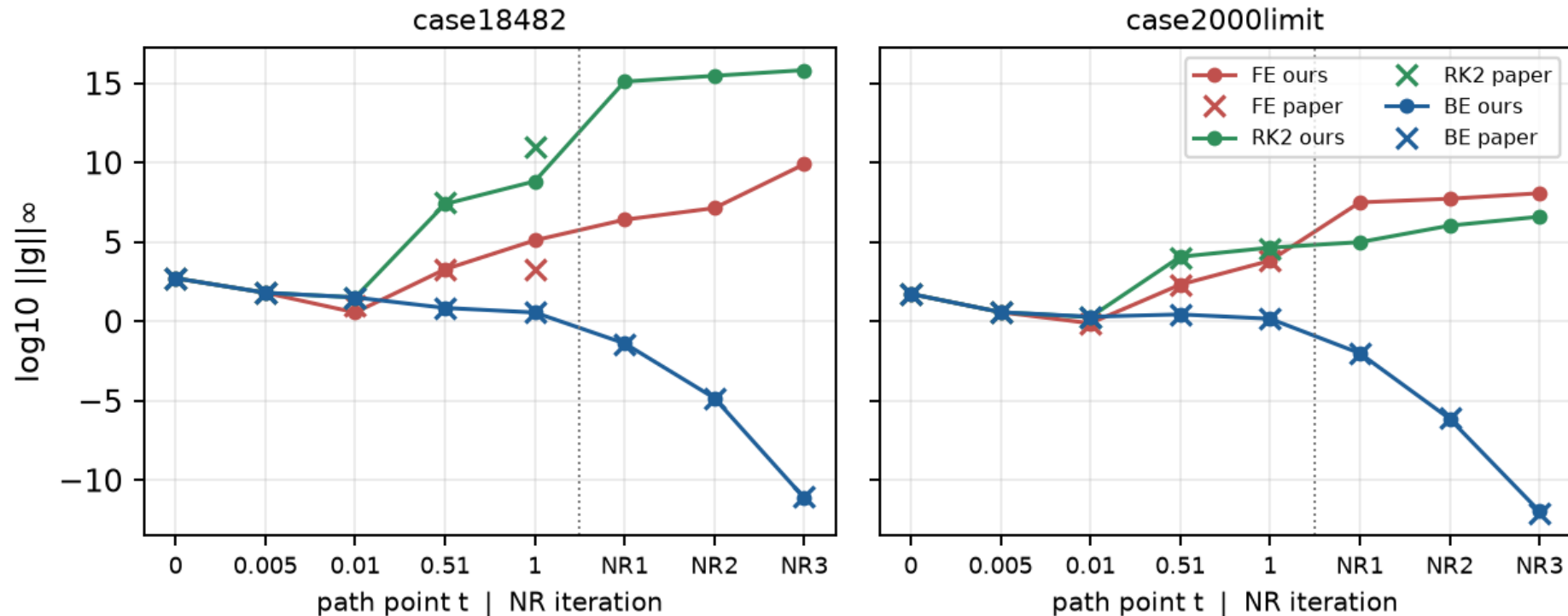
BE is a Newton step

$$\Delta x_{BE} = -G_x(x_k, t_{k+1})^{-1} G(x_k, t_{k+1})$$

- because G is linear in t and $G(x_k, t_k) = 0$
- **one Newton step on the next problem**

Table 5 steps grow slowly (ratio ≤ 6), so there FE survives.

Tables 2 to 4: reproduced



Newton converges after: **FE 3 of 9** · **RK2 2 of 9** · **BE 9 of 9** — the same as the paper

How close is the reproduction?

246 / 255

table cells within a factor of 2 of the paper (96%)

199 / 255

within 10%, as close as two printed digits allow

Outliers: values of runs that already diverged, and one cell (9.7 vs 3.7) that looks like a typo.

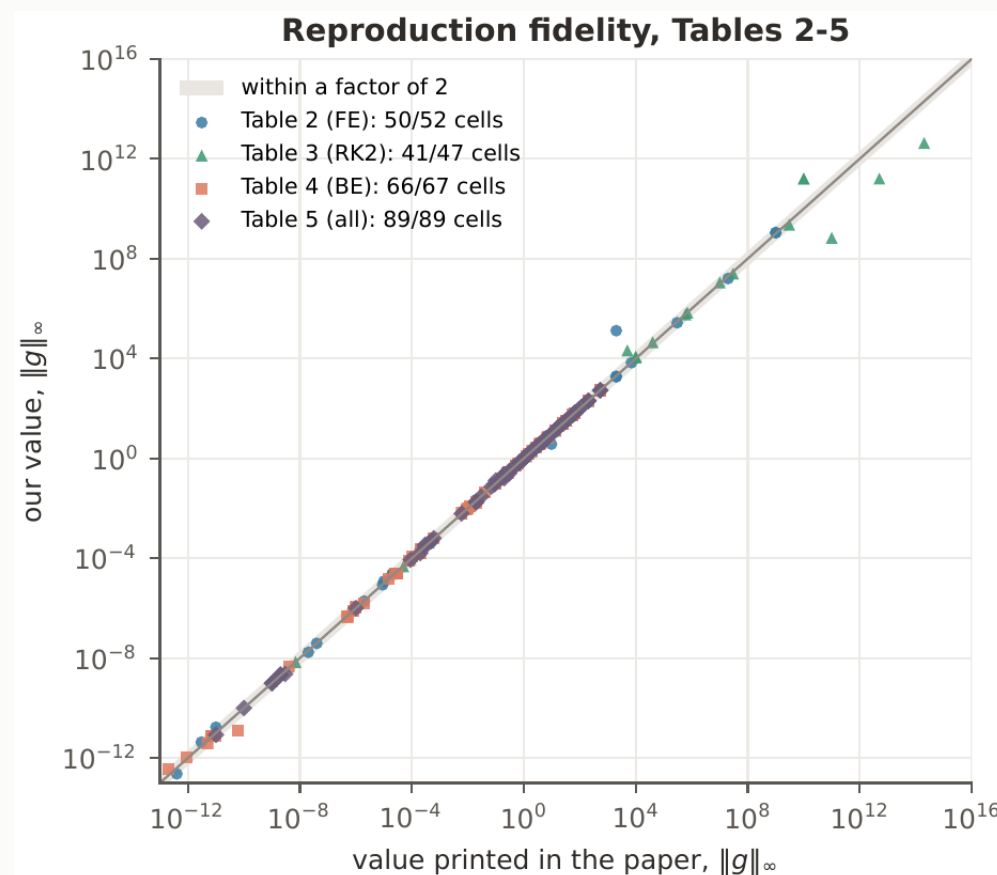
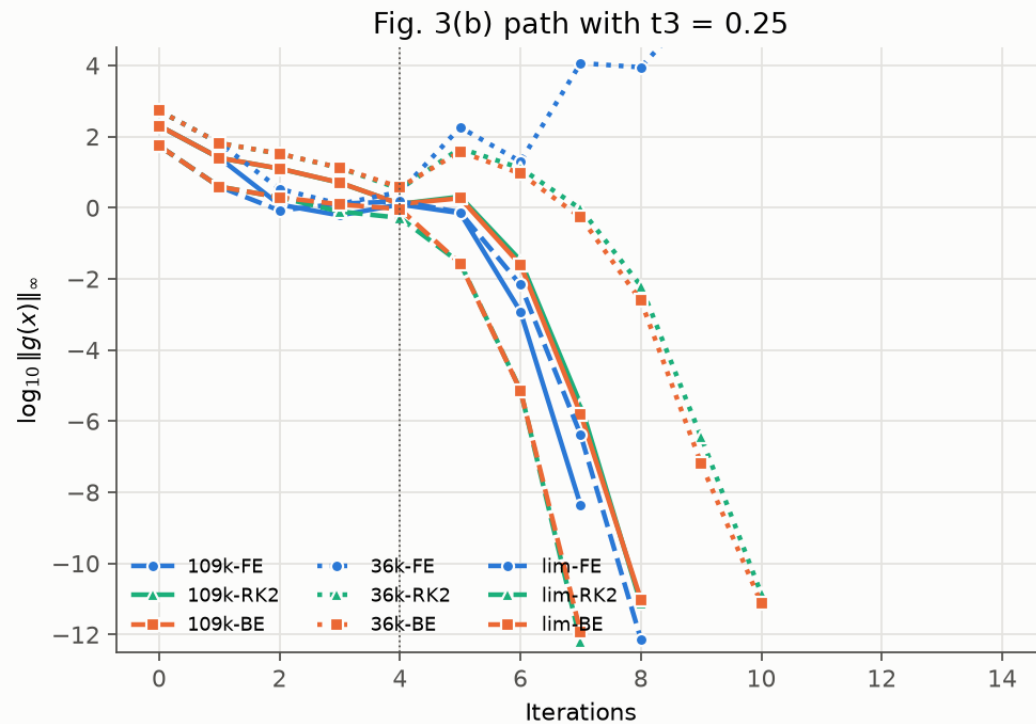


Figure 3 and Table 6: five-point paths



case109272, % of NR-MAT time

$t_3 = 0.20$	Ours	Paper
FE	143%	146%
RK2	225%	230%
BE	163%	171%

- paper's failure reproduced (FE, $t_3 = 0.25$)
- median of 7 interleaved runs

A better integrator? RK4 and BE-chord

Step rule (fixed steps)	Solved of 170	Why
Backward Euler (paper)	127	
BE-chord (3 iterations, 1 LU)	89	iteration does not contract for $\Delta t \approx 1$
RK4	13	explicit: every stage overshoots

Higher order does not help when the step is outside the stability region.

A fair test: every rule picks its own steps

Mean LUs per run with Richardson step control (with Sadman)



BE is still

3.5× to 8×

cheaper

The ODE is stiff: implicit beats high order.

Golden-section search



$$c = b - r(b - a), \quad d = a + r(b - a), \quad r = \frac{\sqrt{5}-1}{2} \approx 0.618$$

- keep [a, d] if $f(c) \leq f(d)$, else [c, b]
- one point is reused: one new solve per step

Minimise

number of LUs of the whole solve

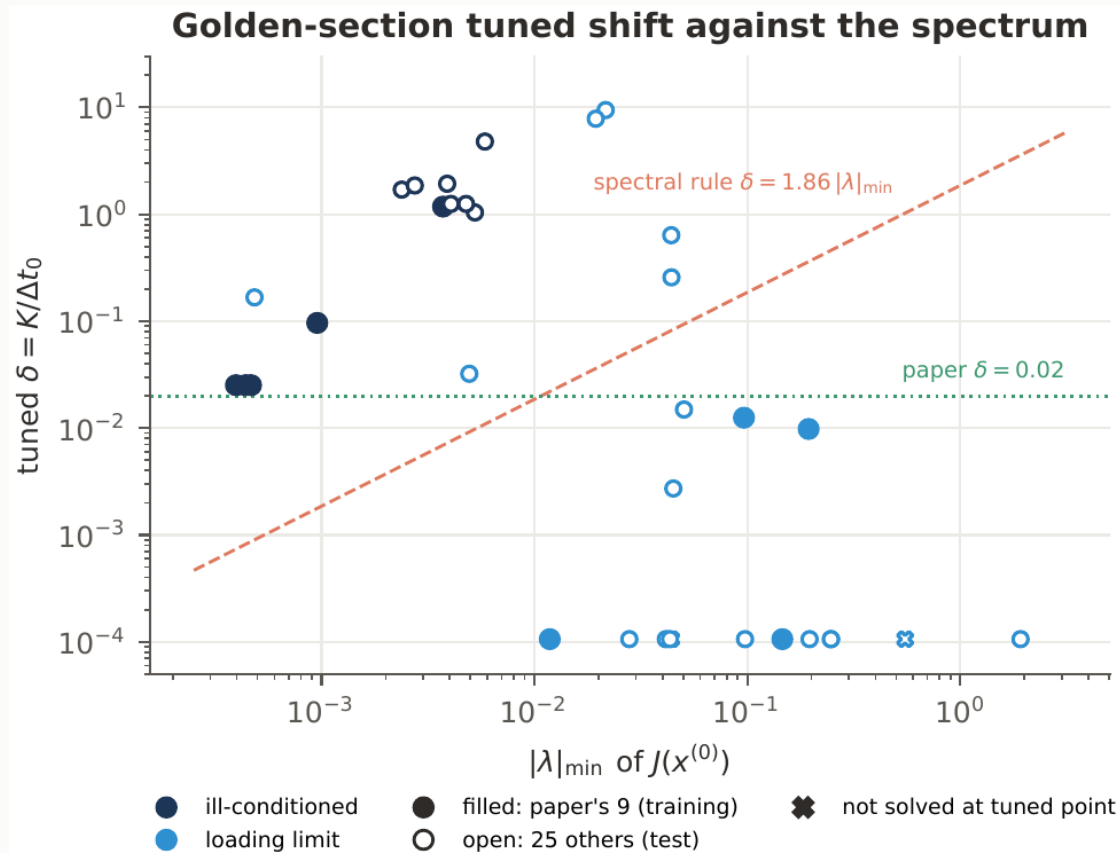
Over

$\log_{10} \delta \in [-4, 1]$, then $\log_{10} \Delta t_0$

Cost

11 + 10 = 21 solves per grid

What tuning δ showed



Rules fitted on 9 grids, tested on 25

How δ is chosen	Test solved
paper, $\delta = 0.02$	19 / 25
fixed, $\delta = 0.011$	19 / 25
spectral, $1.86 \lambda _{\min}$	20 / 25
golden search	23 / 25

- best δ : 0.025 to 4.8 on ill-conditioned grids
- search wins, but costs 21 solves

Summary

- Tables 2 to 6 and Figure 3 reproduced: 96% of cells within 2× of the paper
- FE overshoots by t_{k+1} / t_k ; BE is a Newton step on the next problem
- RK4 and BE-chord do worse; under error control BE is still the cheapest
- The best δ differs per grid, and no simple rule predicts it

Thank you. Questions?